

Using Image Recognition Technology to Assist Visually Impaired People as they Cross the Road

Author: Jiang Yu Hsuan, Lin Ni Ko, Su Zi Han

Supervisor: Huang Jian Ting

Motivation

We want to reduce inconvenience in the visually impaired lives.

Purpose

This research aims to enhance the road safety of visually impaired individuals through the use of an assistive tool. The objective is to develop a machine learning program that can analyze images and identify traffic lights and crosswalks. Once the program successfully recognizes these elements, it can be further extended to provide notifications to visually impaired individuals, thereby offering assistance in their navigation.

Result

We have successfully imported, labeled, and trained image data, allowing the system to recognize whether there is a crosswalk and whether the traffic light in front of the camera is red or green. The following are the training results in terms of precision (P) and recall (R), listed from top to bottom: average, redlight, greenlight, zebraline. From the graph, it is evident that the precision and recall for zebralines (crosswalk) are relatively low.

P	R
0.895	0.841
0.972	1
0.99	1
0.723	0.522

Conclusion

Different colors of crosswalks can indeed impact the recognition results and data. Therefore, we aim to collect more photos for testing purposes, considering the variations in crosswalk colors.

In the case of advanced cameras, capturing images of green lights may cause flickering issues that can also affect the recognition results. This potential problem needs to be addressed.

Traffic light patterns can vary in different regions. To account for these variations, we intend to classify different styles of traffic lights before labeling them accordingly. This categorization will aid in improving the accuracy of the recognition system.